

Canopy temperature algorithms in simplace

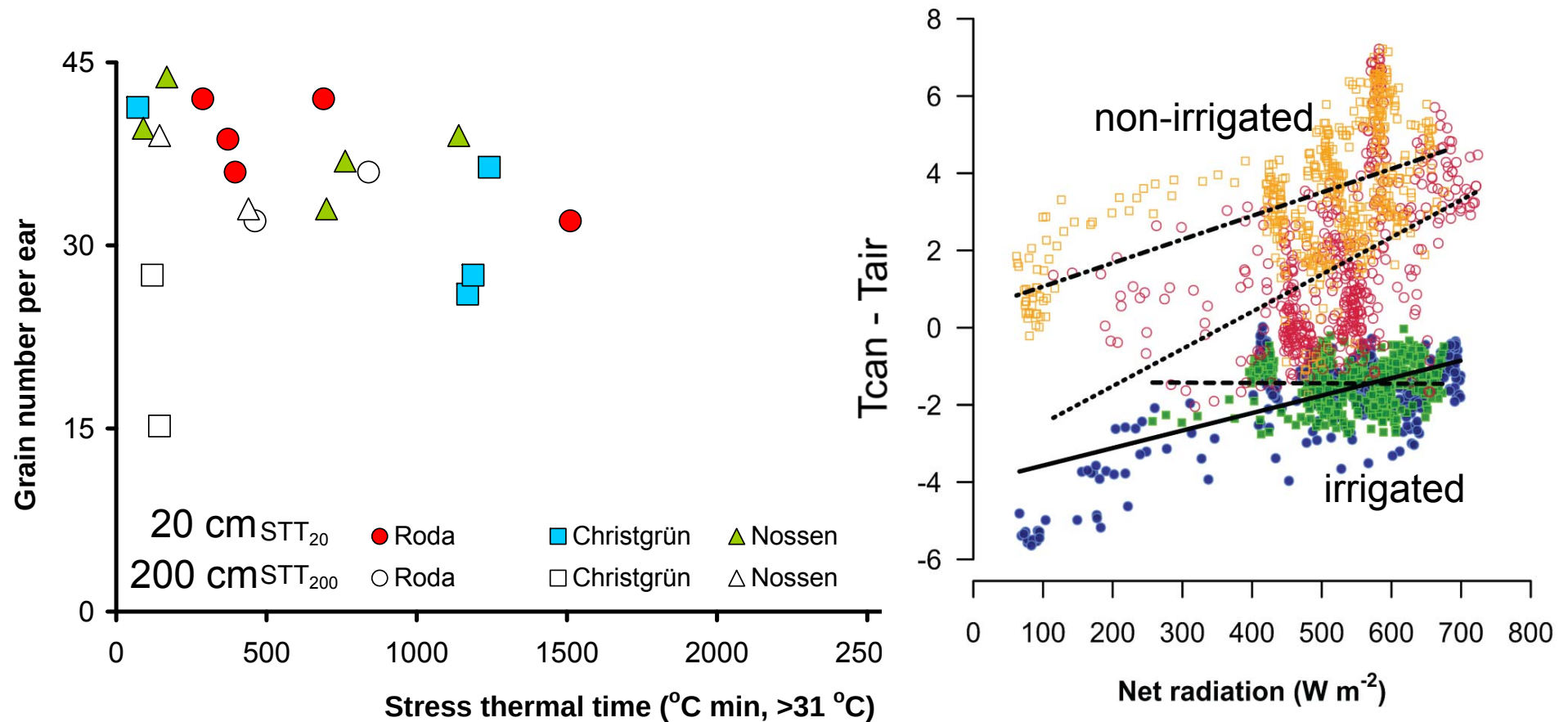
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Our approach
First evaluation results

Summary, open questions and challenges

- Which processes should canopy temperature be used with?
- What part of the crop experiences “canopy temperature”?
- How to parameterize hourly stress thermal time?
- Method sensitive to low wind speeds
- Radiometric vs aerodynamic surface temperatures

How important is consideration of canopy temperature for heat stress?



(Siebert, Ewert, Eyshi Rezaei et al., 2014)

Canopy temperature, T_c :

$$T_c = \underbrace{T_a}_{\text{Air temp}} + \underbrace{\frac{(R_n - G)r_a}{\rho C_p}}_{\text{Radiative heating}} \frac{\gamma^*}{\Delta + \gamma^*} - \underbrace{\frac{VPD}{\Delta + \gamma^*}}_{\text{Evaporative cooling}}$$

$$\gamma^* = \gamma \left(1 + \frac{r_c}{r_a} \right) \quad \text{Canopy resistance - complex}$$

(Clawson et al., 1989)

Simplifying assumptions:

Full water stress: $r_c = \infty$, upper limit to canopy

No water stress: $r_c = \text{constant}$, lower limit to canopy temp

where T_a is 2m air temp; R_n net radiation; G soil heat flux; VPD vapour pressure deficit; Δ slope SVP; ρ air density; C_p specific heat of air; γ psychrometric constant; r_c canopy resistance; and r_a aerodynamic resistance

simplified approach in simplace:

$$T_c = T_{c,lower} + (1 - K_s)(T_{c,upper} - T_{c,lower})$$

where K_s crop water stress

complete water stress, $r_c = \infty$

$$T_{c,upper} = T_a + \frac{R_n r_a}{\rho C_p}$$

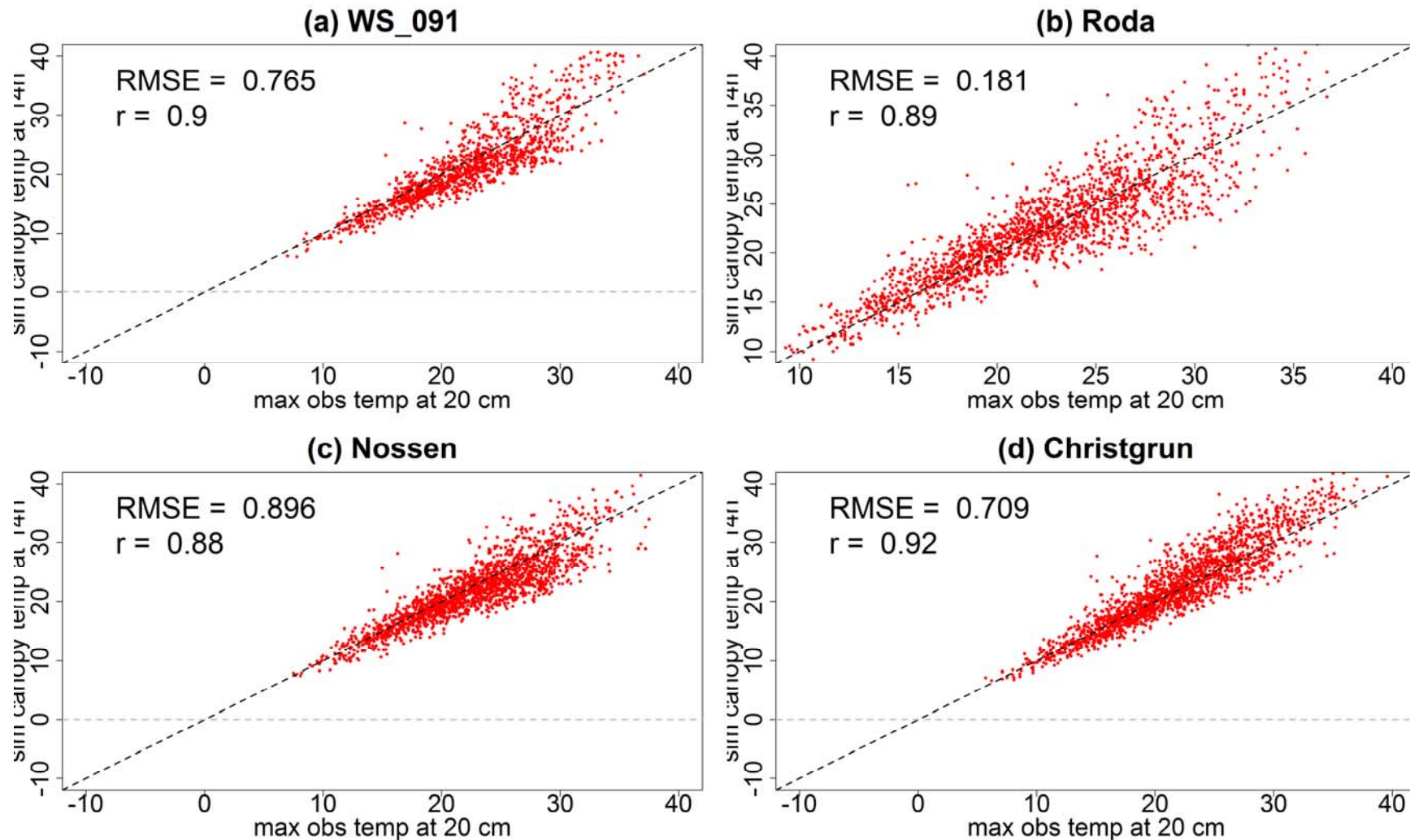
no water stress, $r_c = \text{constant}$

$$T_{c,lower} = T_a + \frac{R_n r_a}{\rho C_p} \frac{\gamma^*}{\Delta + \gamma^*} - \frac{VPD}{\Delta + \gamma^*}$$

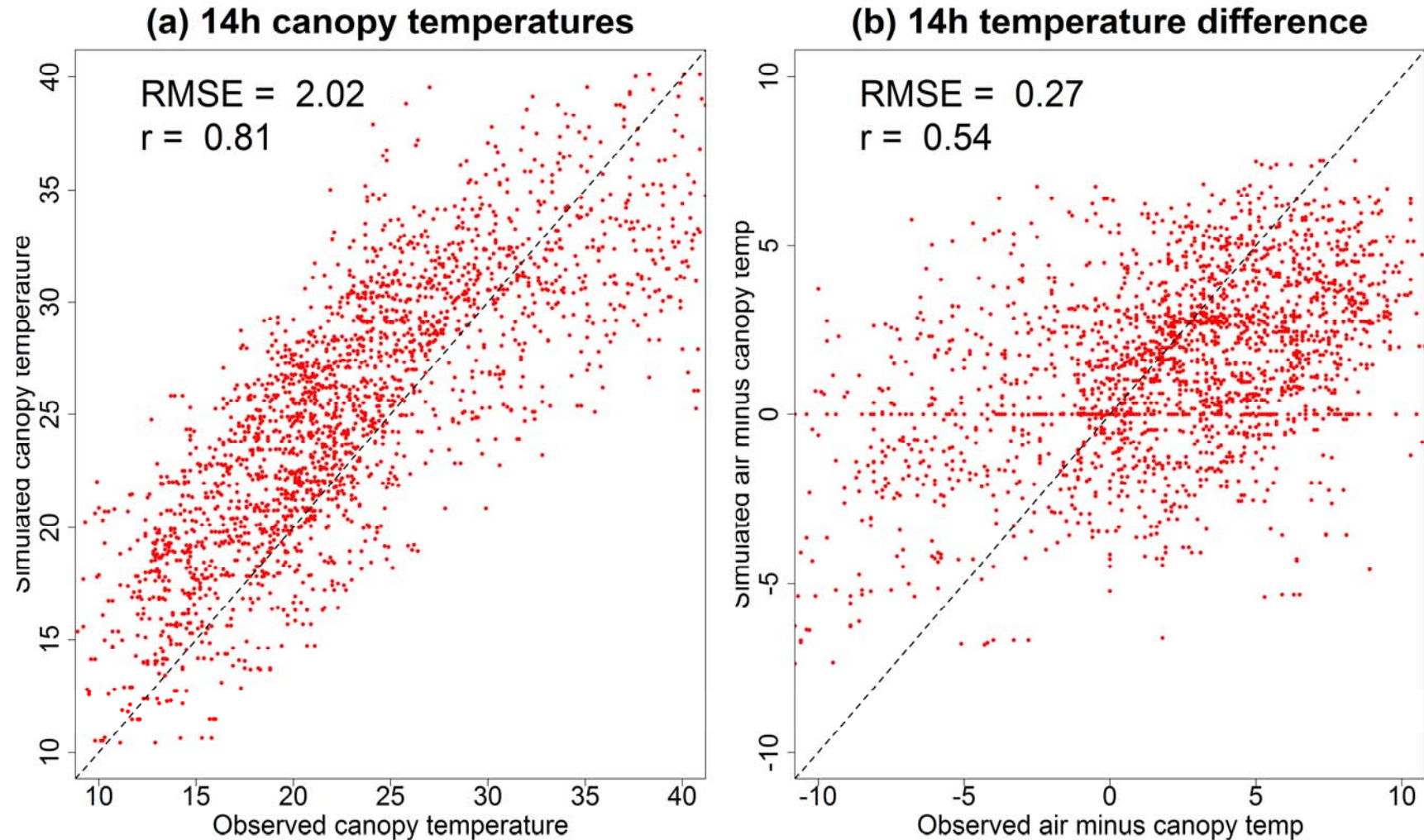
Approach, assumptions, boundary condition, details

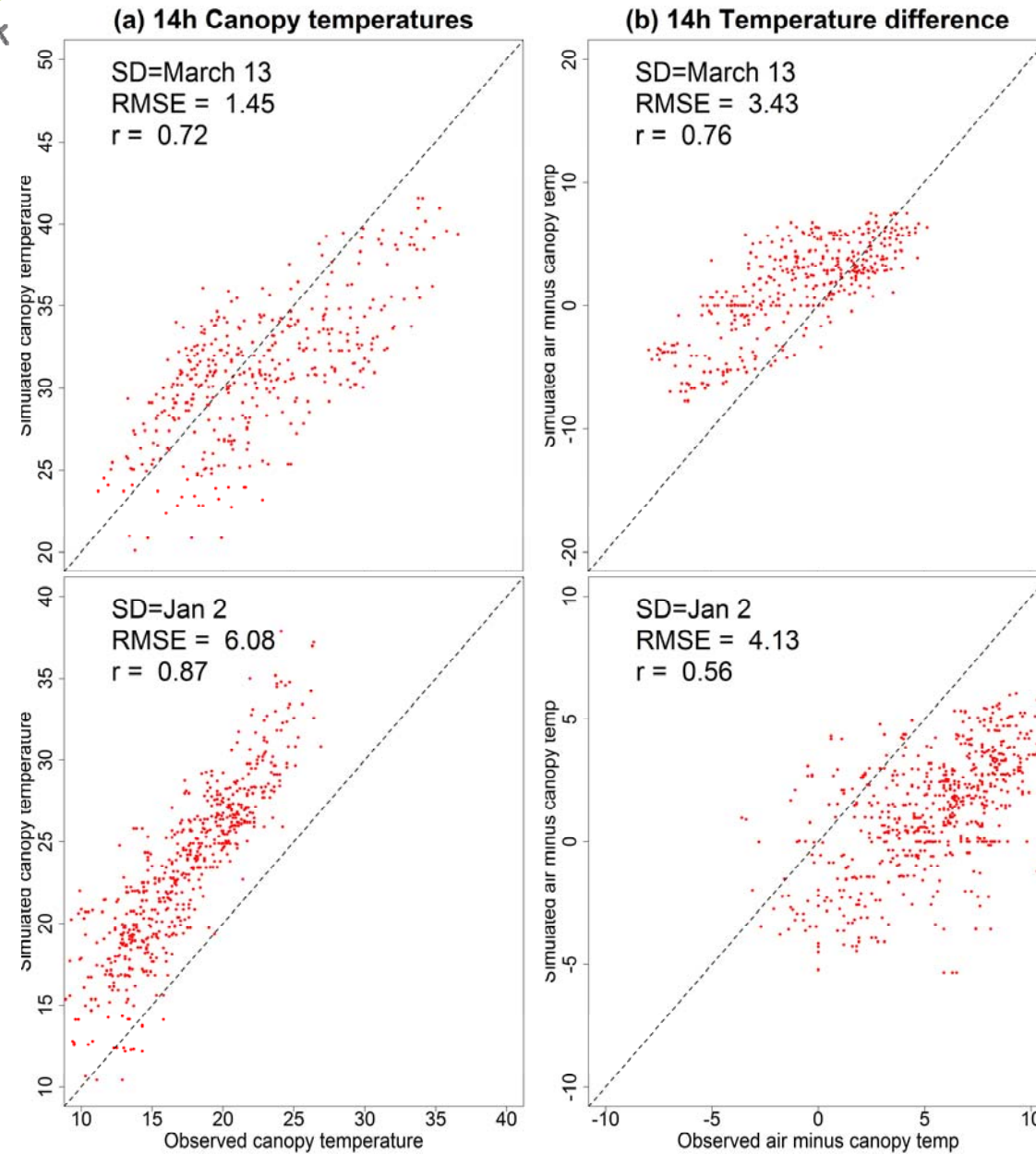
- Hourly timestep
- generate hourly temperature & radiation distribution
- $r_a = f(\text{wind}, \text{height}, \text{LAI}, \text{stability regime})$
- iterate for “exact” Monin-Obukhov length (Thom, 1975)
- hourly water stress

Evaluation over grass (Germany, 2000-11)

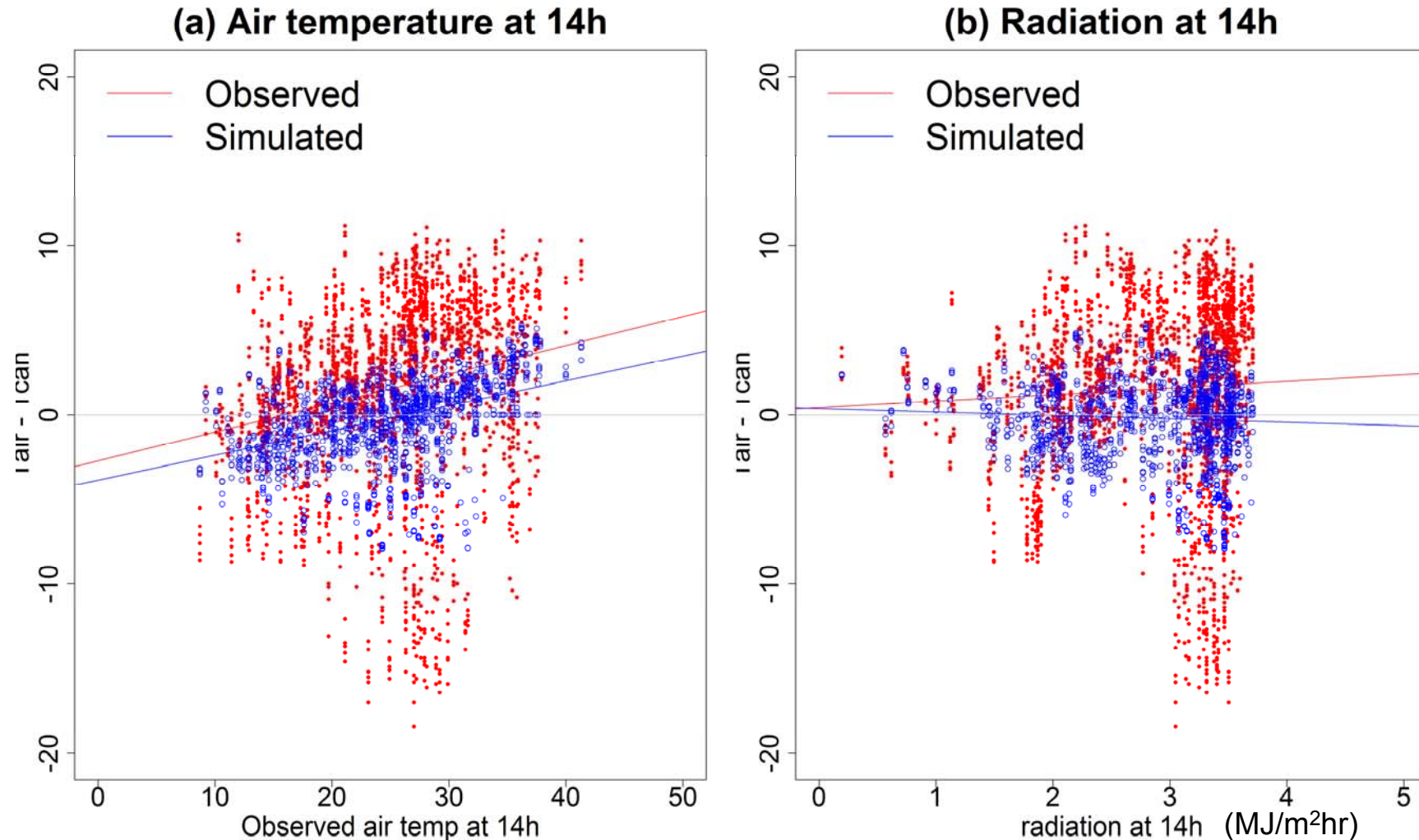


Evaluation with HSC canopy temperatures

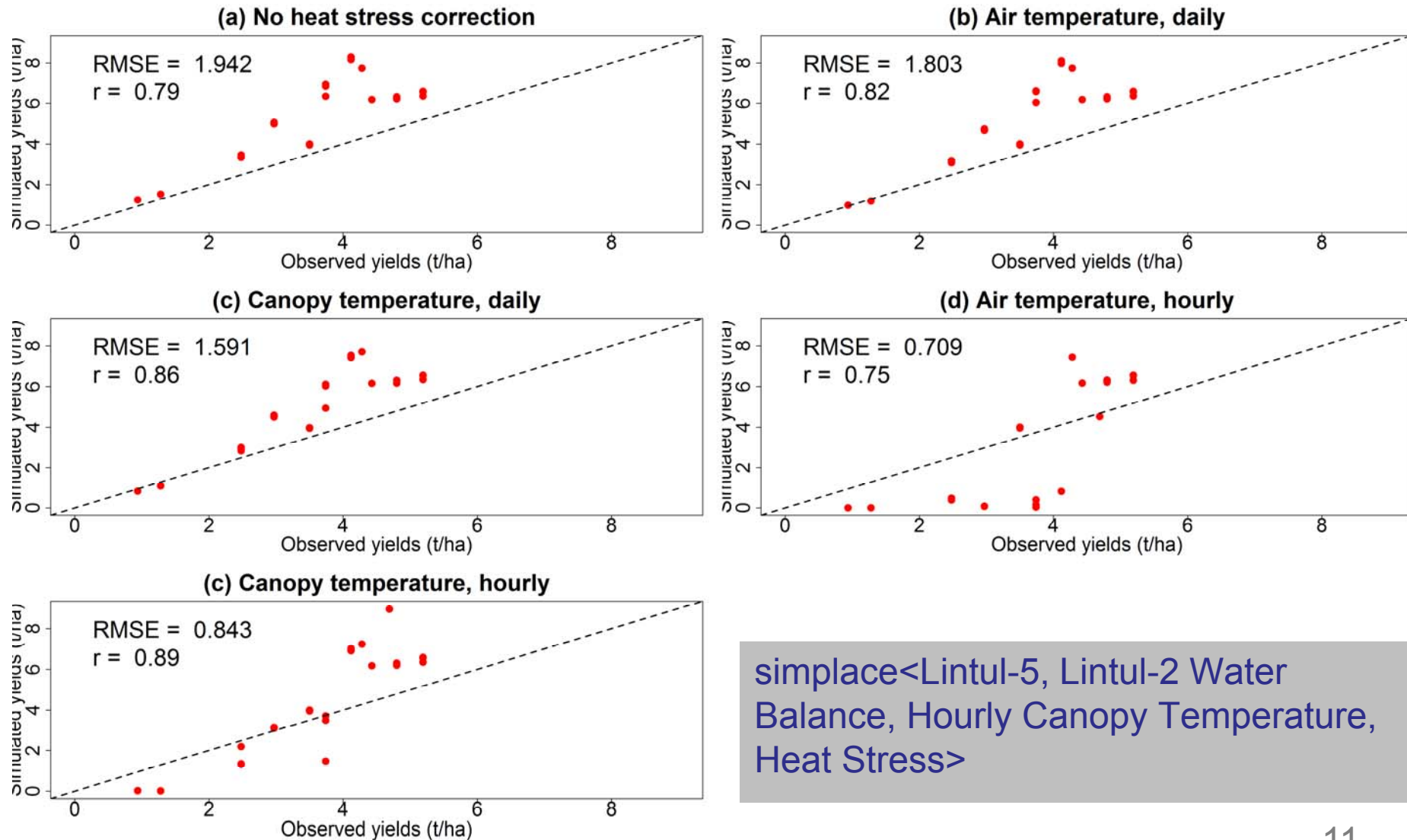




What explains our “poor” simulation of $T_{\text{air}} - T_{\text{can}}$?



Does canopy temp improve yield simulation?



simplace<Lintul-5, Lintul-2 Water
Balance, Hourly Canopy Temperature,
Heat Stress>

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